RAG Assignment: Baseline (Bi-Encoder) vs. Re-Ranking (Cross-Encoder)

Document: Digital Personal Data Protection Act, 2023 (India).
Setup: 92 chunks (one per section, long sections split with overlap, the Schedule split per row). Bi-encoder `multi-qa-MiniLM-L6-cos-v1`, cross-encoder `ms-marco-MiniLM-L-6-v2`, Gemini for answers (temperature 0). Both pipelines use the same chunks and the same prompt. Everything ran on a CPU-only Colab runtime.

1 Architecture comparison

Bi-encoder

- Encodes the query and each chunk separately into one vector each. Similarity is the cosine of the two.
- Chunk vectors are computed once and stored, so search is cheap.
- The query and chunk never interact, so negation, exceptions and word-to-word relationships get compressed into one vector.

Cross-encoder

- Takes the query and chunk together as one input, with self-attention across all their tokens, and outputs one relevance score.
- More accurate, because it compares query words directly against chunk words.
- Nothing can be precomputed. Every (query, chunk) pair needs a forward pass at query time.

2 Rank shifts and answers

25 queries were screened (20 original plus 5 extra negation). Two moved from outside the baseline Top 3 into the re-ranked Top 3.

Query	Gold chunk	Baseline	Re-ranked
Is notice and consent required when personal data is processed only for research or statistical purposes?	Section 17(2)(b), chunk 43	8	3
What is the fine for not telling the Board about a personal data breach?	Schedule penalty row 2, chunk 86	10	1

Query 1 (exception case)

- The baseline Top 3 was all general consent and notice chunks (Sections 6, 5, 5). The exemption in 17(2)(b) was buried at rank 8.
- Answer 1 said “Not found in the provided context.”

- Answer 2 correctly said no, citing 17(2)(b): research, archiving and statistical processing is exempt if it is not used for decisions about a specific Data Principal and follows the prescribed standards.

Query 2 (vocabulary mismatch)

- The query says "fine" and "telling the Board", while the Act says "penalty" and "notice of a personal data breach".
- The baseline pulled Board-related chunks (Sections 27 and 28). The Schedule row only reached rank 10.
- Answer 1 again said "Not found". Answer 2 gave the correct figure, up to two hundred crore rupees, citing the Schedule row and Section 8(6).
- I call this a vocabulary mismatch rather than a strict keyword trap, since the problem was different wording, not an unrelated passage matching on shared words.

Baseline answer quality

- The baseline never hallucinated. In both cases it omitted the answer because the key chunk sat outside the Top 3.

Negation

- I tested 9 negation-style queries. None showed a negation case that re-ranking restores, and the cross-encoder never moved one into the Top 3.
- The four original queries all had the gold chunk at rank 1 in both pipelines.
- Five harder queries targeted prohibition and immunity clauses ("no suit shall lie", "shall be invalid", "no civil court shall have jurisdiction"). Three of them had the gold chunk at rank 1, 1 and 2.
- **Section 39 (civil court bar):** "Can a Data Principal go to a civil court instead of the Board…" had the gold chunk at rank 9 in both pipelines. This is a real baseline failure that re-ranking did not fix. A likely reason (untested) is that the cross-encoder is trained on web passage relevance, which does not reward a short clause that only says what is not allowed.
- **"NOT required to give notice" query:** gold moved from rank 5 to rank 7 after re-ranking, so re-ranking made it worse. Neither pipeline reached the Top 3.

Retrieval misses (re-ranking cannot help)

- The national-security exemption, the "child" definition and the Section 11(2) query never had the gold chunk in the Top 10.
- The cross-encoder only re-orders the 10 candidates it is given.
- The Section 11(2) miss may partly be a labelling artifact. I did not check whether the key phrase was split across a chunk boundary.

Gold ranks use the section, plus a key phrase for some queries, so they are a reasonable but not perfect measure.

3 Latency profiling

CPU only, mean over the original 20 queries × 5 runs, LLM time excluded.

Stage	Mean (ms)	Std (ms)	p95 (ms)
Pipeline 1: baseline retrieval	28.14	15.76	61.40
Pipeline 2: + cross-encoder re-rank	1747.35	846.15	3601.72
Pipeline 2 total	1775.49	854.13	3635.33

- Re-ranking only 10 chunks adds about 62× the baseline retrieval time.
- A GPU would shrink this a lot, but the ratio would still be large.

4 Why not run the cross-encoder over everything?

- A cross-encoder's score depends on the query, so chunk representations cannot be precomputed.
- Every query needs a forward pass for every chunk, so cost grows linearly with corpus size, per query.
- I measured 11.7 s for all 92 chunks, about 127 ms per pair. That extrapolates to roughly 2.1 minutes per query for 1,000 chunks and about 212 minutes for 100,000.
- A bi-encoder embeds documents once offline. A query costs one encoding plus a vector search (about 19 ms here), which approximate indexes make faster still.
- The standard design is a fast bi-encoder to shortlist candidates and a slow, accurate cross-encoder to reorder only that shortlist.